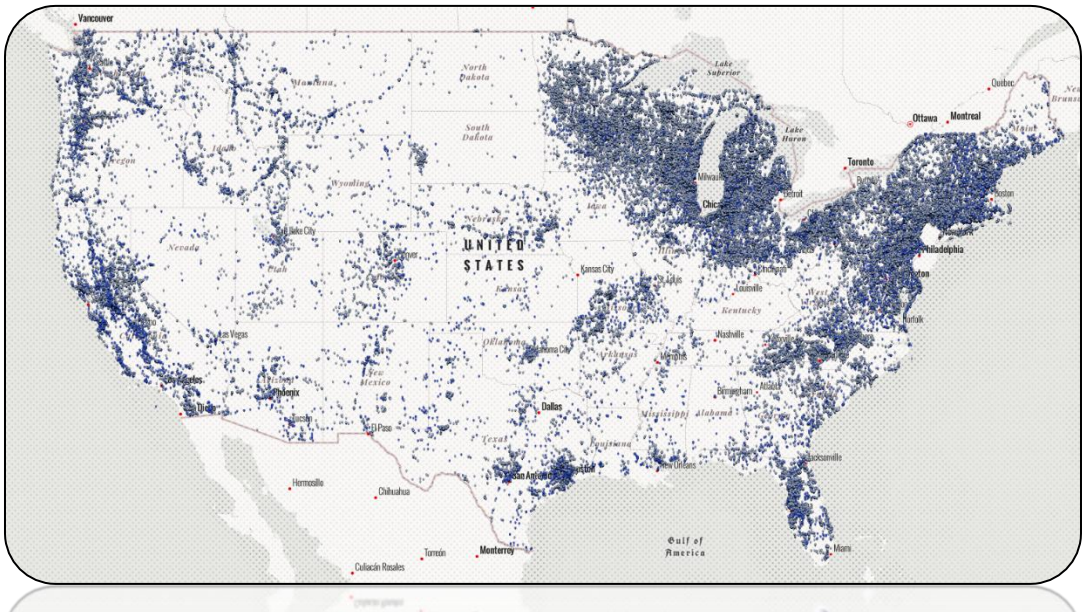
United States
Environmental Protection
Agency

Transient and Non-Transient Non-Community Water System Service Areas

Methods Document



Acronym List

CWS	Community Water System
ECHO	Enforcement and Compliance History Online
NCWS	Non-Community Water System
NTNCWS	Non-Transient Non-Community Water System
PWS	Public Water Systems
SDWIS	Safe Drinking Water Information System
TNCWS	Transient Non-Community Water System

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Links to Dataset and Further Information

The purpose of this document is to describe how the Transient and Non-Transient Non-Community Water System Service Area dataset was developed.

For further information on service area datasets for all public water systems, visit: https://www.epa.gov/ground-water-and-drinking-water/community-water-system-service-area-boundaries

The Non-Community Water System Service Area dataset REST service can be found on ArcGIS Online: https://services.arcgis.com/cJ9YHowT8TU7DUyn/arcgis/rest/services/Transient_and_Non_Transient_Non_Community_Water_System_Service_Areas/FeatureServer

The Public Water System Service Area Application, for easy viewing of the data, can be found here: gispub.epa.gov/serviceareas

And finally, visit our GitHub site for detailed information on public water system service areas:

https://github.com/USEPA/ORD_SAB_Model

Public Water System Definitions

EPA defines three types of public water systems. The dataset developed and explained in this document contains two of the three: Transient and Non-Transient Non-Community Water Systems.

- ***Non-Transient Non-Community Water System*** **(NTNCWS)**: A public water system that regularly supplies water to at least 25 of the same people at least six months per year. Some examples are schools, factories, office buildings, and hospitals which have their own water systems.
- ***Transient Non-Community Water System*** **(TNCWS)**: A public water system that provides water in a place such as a gas station or campground where people do not remain for long periods of time.

The third type are Community Water Systems. These service areas have also been published by EPA's Office of Water and are defined by:

- ***Community Water System*** **(CWS):** A public water system that supplies water to the same population year-round.

Methods

Parcel Geographies

The purpose of this effort is to delineate a geography for all Transient and Non-Transient Non-Community Water Systems (NCWSs). These systems are typically located at a single facility, such as a business, hospital, factory, or rest stop. At this geographic scale, assigning a parcel geometry to each NCWS is the most appropriate approach. Parcels are typically managed by counties and delineate properties for property tax purposes. Census geographies—like blocks and block groups that were used in the CWS service area dataset—are too large to conform with this infrastructure. Parcels are an appropriate geographic scale for these types of water systems and are available for all US states and its territories, except Guam.

The parcel data used for the NCWS service areas are sourced from Regrid's[1] 2025 release. Parcels were downloaded and merged by state or territory. Any duplicate parcel geometries were deleted. Guam was the only state or territory where no parcels were available for use and thus no NCWS service areas were delineated for Guam.

Public Water System Data

Attribute and geographic data were compiled from the 2025 (Quarter 3) Safe Drinking Water Information System (SDWIS) download available from EPAs Enforcement and Compliance History Online (ECHO) database. All public water systems that were active (A) and listed as a 'TNCWS' or 'NTNCWS' were selected.

There is no definitive data source as to where NCWS are located. However, there are clues that point to these systems' locations. The following 'clues' were used to help locate them:

- **Public water system name**
  This information is collected within SDWIS. It is simply the name of the public water system. Many of the names are names of business that can be geolocated with a geocoder. Similar to entering a business name in google maps, 'Vinny's Pizza on Vine' for example, the exact location of the business (or NCWS) can be determined.
- **Primacy agency**
  This information is collected within SDWIS. It is the state, territory, or EPA region which holds SDWA primary enforcement responsibility over the NCWS.
- **Mailing address:**
  This information is collected within SDWIS. The system's mailing address is sometimes, but not always the physical address.
- **Locations of SDWIS facility infrastructure**
  This is non-public geographic information on the locations of public water system facility infrastructure including surface intakes, wells, treatment plants and distribution systems.

The final dataset is divided into two types of location confidence: 'Higher' and 'Lower'. These descriptors can be found in the feature's attribute table. These are qualitative descriptors that are relative to each other. On average, the 'Higher' confidence Transient or Non-Transient Non-Community Water System service areas will more often be correctly located than the 'Lower' confidence systems. However, the

[1] https://regrid.com/

majority of both types are correctly located. Below describes the locational methods employed that determined the 'Higher' and 'Lower' confidence label methods.

'Higher' Confidence Service Area Location Method

Higher confidence location methods for defining the PWS service area leverage converging lines of evidence to map their geographic extent. In other words, multiple pieces of information must point to the same service area location, or parcel, in order for the feature to be considered 'Higher' confidence. The following list describes the geographic evidence that was used to map 'Higher' confidence NCWS boundaries and how these data were curated.

SDWIS Reported Mailing Address

This is the address the public water system provides in SDWIS reporting. There are 93,558 NCWS locations. These addresses are often mailing addresses that can be different from the system's physical address (the location we are trying to map). Unfortunately, we do not know what type of address was provided. For this data to be useful, we need converging lines of evidence to confirm a correct physical location. Also noteworthy: approximately 25,000 TNCWSs and NTNCWS had no address line information (27%).

Before geocoding these addresses, some pre-processing needed to be done. First, duplicate addresses were identified and selected. If there were identical addresses for two or more TNCWS or NTNCWSs, then we assumed that those addresses were mailing addresses and not physical addresses because we shouldn't have the same address for multiple physical locations. Second, if the mailing address state is different from the PWS primacy agency we know the provided address is not the physical address. For these systems, the street address, city, and zip code information was removed, leaving just the system name and state. For the NCWS where the address state was different from the primacy agency, the state code was changed to the primacy agency state code. All locations were then geocoded using esri's world geocoder. The 'PWS Name' was used as address field one and the 'PWS Address' was used for address field two. This geocoder can use business names to match to a "point of interest" location, such as a business, campground, or administrative building, for example. NCWSs with duplicate addresses or systems that didn't provide a mailing address were only geocoded on the system's name and state. The geocoder was run iteratively, state-by-state. After the geocoder was run, unmatched water systems or systems that were not matched to a point address or point of interest location were rerun through the geocoder using just the PWS Name and PWS State (for NCWS that had an address in the first place). When completed, only 'point of interest' matches were added back to the first geocode results. This was done to fix any locations that were potentially using a mailing address and were unmatched or coarsely matched, so that the second round of geocoding was then able to match just the business name without the potentially incorrect mailing address information.

SDWIS Facility Types

These point features are internal to the EPA and represent the locations of unique facility type infrastructure associated with public water systems. This list of SDWIS features is available in Table 1.

Table 1 Unique types of SDWIS point location infrastructure

SDWIS Facility Types
Common Headers
Cistern
Clear Well
Distribution System/Zone
Non-piped, non-purchased
Other
Pressure Control
Pump Facility Impoundment
Sampling Station
Storage
Transmission Main (Manifold)
Well Head
Consecutive Connection
Infiltration Gallery
Intake
Non-piped
Roof Catchment
Reservoir
Spring
Well (Source)
Treatment Plant

For NCWS's, the most common facility types are wells. A water system can have multiple facility types as well as multiple single types—such as having multiple wells for one water system. These facility locations provide geographic clues to the physical location of the public water system service area. Some of the locations are very accurate and some can be miles from where the actual system is located. Generally, the locations are fairly close to the actual water system location. There are 149,000+ unique facility type locations for the 93,000+ active NCWSs.

Parcel Owner Name

The last type of data we used to locate our NCWSs is parcel owner name information from Regrid. This data was used to validate the SDWIS facility infrastructure and geocoded address locations. Often the public water system name is similar to their associated parcel owner's name.

Joining SDWIS locations to Parcels

After all county parcel data was merged to its associated state, the SDWIS addresses were geocoded, and all NCWSs' SDWIS facility infrastructure data was curated for that state, the data were spatially joined to parcels. A one-to-many spatial join (parcels to geocoded addresses and infrastructure point data) was performed and only parcels which intersected these points were returned. For intersecting features, the parcel owner name and unique parcel ID were identified. The PWS Name and PWSID for intersecting

geocoded features and SDWIS facilities were joined to the parcel. The count of intersecting geocoded addresses and the count of intersecting SDWIS facilities was summed to each parcel.

The next step was to create a unique ID (UID) from the output of the spatial join operation by concatenating the PWSID and unique parcel number. The spatially joined data was then aggregated by the sum of the count of SDWIS facility infrastructure and sum of the count of geocoded addresses, by the newly created UID. We identified UIDs with multiple lines of evidence that the parcel is correctly identified with the correct PWSID. We looked for UID's that had both a geocoded location and at least one SDWIS facility location. These UIDs were then given a label of 'BOTH'. A parcel is likely correctly identified to its correct public water system if it has both a geocoded location and SDWIS infrastructure with the PWSID intersecting the same parcel. Next, we identified UIDs that have at least two SDWIS facility locations. These UIDs are labeled with 'SDWIS'. A parcel is likely correctly identified if it has more than one SDWIS location of the same PWSID.

Next, we conducted a fuzzy name matching exercise between the parcel owner name and the PWS name. The formula calculates the similarity percentage between two strings using a Levenshtein Distance formulation. The VBA code for this function can be found on GitHub.[2] The rationale: if the parcel owner name is similar to the PWS name, it is likely correctly located. An example: PWS Name = LAKESIDE MENNONITE SCHOOL and the parcel owner name = CHURCH LAKESIDE MENNONITE. This received a Levenstein score of .44, which meets the threshold needed to be considered a possible match. Roughly, a score of .3 and above was considered a possible match, and then all possible matches were manually reviewed and verified as logical joins. UIDs that had a name match were labeled with 'NAME'. Table 2 describes the three different match types, for reference.

Table 2 The three different match types used to connect a PWS to its correct parcel

Match Type	Description
BOTH	The PWS's geocoded mailing address and at least one SDWIS facility location were both located on the same parcel
SDWIS	Two or more SDWIS facility locations for the same PWS fell on the same parcel
NAME	The PWS's name from either a SDWIS facility or a geocoded mailing address that was joined to a parcel is similar to the parcel owner's name

After the three match types were calculated, NCWS were checked for having multiple contradictory match locations. To select the most probable parcel for a given NCWS, we determined which matching outcome(s) are most robust for a NCWS with multiple parcel options. To do so, we concatenated the

[2] vba_fuzzymatching, Kyle Deer, https://github.com/kyledeer-32/vba_fuzzymatching/blob/main/README.md

match types for each UID—'BOTH', 'SDWIS', and 'NAME'. The type and count of matches determine the strength of the converging lines of evidence for the confidence we have for a parcel representing the true public water system location. There are seven unique concatenated permutations: BOTH; NAME; SDWIS; BOTH & NAME; NAME & SDWIS; BOTH & SDWIS; and NAME & SDWIS & BOTH. We then qualitatively rank the confidence level for each permutation type, from 5-1—with higher numbers suggesting more confidence in the correct location.

Table 3 The unique permutations of match types and their associated rank based on locational confidence

Match Type	Rank
BOTH & SDWIS & NAME	5
BOTH & NAME	4
BOTH & SDWIS	3
SDWIS & NAME	3
BOTH	2
NAME	2
SDWIS	1

Duplicate PWSID rows were then deleted by their rank, shown in Table 3. Duplicate PWSID rows with a Rank of 0 were deleted first, and Rank 4 was deleted last. This ensures that if there are multiple location options for a single water system, the parcel with the highest locational confidence (and a score greater than 0) ends up in the final dataset. The methods outlined above are given in an example in Table 4 below. It shows 5 UIDs and 4 unique PWSs—how they were matched and ranked given our different methods for matching their locations. Note that rows 1 and 2 have the same PWS, but row 2 would be removed because it has a rank of 0 while row 1 has a rank of 5. In total, 59,981 service areas were matched using this method.

UID (PWSID + Parcel ID)	PWSID	PARCEL NAME	PWS NAME	String Similarity	Sum of Geocoded Count	Sum of SDWIS Count	SDWIS Match	BOTH Match	NAME Match	Match Concat	Rank
ID1050005_RP0045	ID1050005	GLANBIA FOODS INC	GLANBIA FOODS INC	1	1	3	SDWIS	BOTH	NAME	'IS BOTH N	5
ID1050005_RP46N	ID1050005	SIMPLOT FEED LOTS	GLANBIA FOODS INC	0.09	0	1					0
ID1050012_RP46N	ID1050012	JEFFERSON COUNTY	JEFFERSON COUNTY LAKE	0.76	0	2	SDWIS		NAME	DWIS NAM	3
ID1050013_RP46	ID1050013	SHOSHONE COUNTY	GENE SHOSHONE COUNTY	0.52	1	1		BOTH	NAME	OTH NAM	4
ID1050033_RP00	ID1050033	CUSHING FAMILY	GARDEN VALLEY LDS HOUSE	0.26	0	2	SDWIS			SDWIS	1

Figure 1 Example how UIDs are matched to different location methods and how they are ranked.

'Lower' Confidence Service Area Location Method

NCWS that were not matched to a parcel in the 'Higher' confidence location method describe above were selected as a candidate for the 'Lower' confidence location method. This method is more straightforward than the other method. While the 'Higher' location method requires multiple lines of evidence to match a PWS to a parcel, this method is less stringent and only requires a single line of evidence. Here we only use the SDWIS facility locations data. Specifically, we only look at SDWIS locations that have only one point location per PWSID. Using a list of unmatched PWS from the method above, we identify those PWS with a single SDWIS location. That SDWIS location is then spatially joined to its associated parcel and is considered a match. 15,412 PWS were matched using this method.

Validation

Using the dataset created from the methods described above, we validate the efficacy of the method by comparing the output to NCWSs delineated by California's Water Resource Control Board.[3] According to CA's website regarding the creation of the public water system boundaries, "In order to provide an accurate data set of service area boundaries for California drinking water systems, the Division of Drinking Water of the California Water Resources Control Board (SWRCB DDW) has undertaken a project to vet and verify the data collected by the Tracking California's Water Boundary Tool."

This CA dataset contained 1,847 out of the 3,807 NCWSs modeled by the EPA. Of those common systems, we intersected the EPA boundaries with any CA boundaries and compared the intersections containing a common PWSID. Results:

- 1,549/1,847 NCWS with matching boundaries (84%)
- Of 'Higher' confidence locations: 1245/1411 matching boundaries (88%)
- Of 'Lower' confidence locations: 304/436 matching boundaries (70%)

However, just because there wasn't overlap doesn't necessarily mean the modeled boundary is 'wrong'. A NCWS can envelop multiple parcels, and the model just selected one of the correct parcels.

State Sourced Service Areas

The California service areas described in the Validation section above were then incorporated into the dataset. Because these service areas are considered accurate, we replaced any parcel derived service area in California with the corresponding service area derived from California's Water Boundary Tool. In total, 2,077 state sourced California service areas were incorporated into the final dataset.

[3] Feature service can be found here:
https://www.arcgis.com/home/item.html?id=fbba842bf134497c9d611ad506ec48cc

QA/QC

After the matching routines described above are completed for each state and territory, both PWS location method datasets were merged into a national dataset. A 'Location Confidence' field was created in the attribute table and systems were labeled with their associated 'Higher' or 'Lower' match types. The national dataset was then manually reviewed for errors. It was discovered that some PWSs were mapped to extremely large parcels. This typically happened in Western states where a national park's well for their visitor center or a campground well was matched to a parcel owned by the Forest Service or BLM, for example. Figure 2 illustrates some large parcels in Idaho, Nevada, and Utah (blue polygons). These parcels could be over 1,000 sq km in area. While the NCWS would in fact be within the large parcel, it doesn't provide much quantitative or qualitative use to end user (i.e., knowing that somewhere within a 1,000 sq km area there is a public bathroom that services a transient population with water). Because of this, it was decided that any PWS matched to a parcel greater than 26 sq. km. was deleted. Roughly 300 PWS were removed using this criterion.

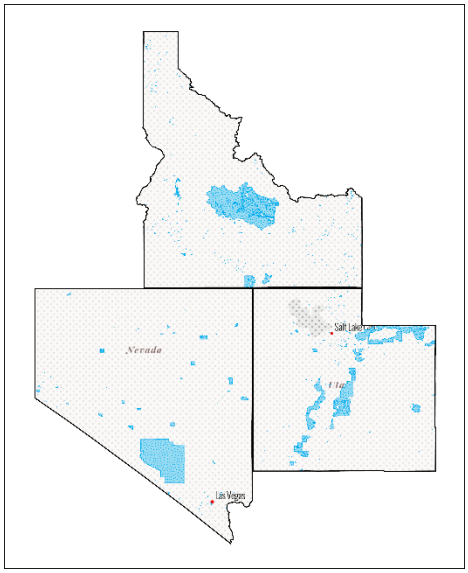


Figure 2 Example of very large parcels being associated with PWSs in three Western States. Blue polygons represent the PWS service area.

It was found that service areas could erroneously be associated with the wrong parcel when a geocoded NCWS address was matched to the system's mailing address (and not the physical address) and the parcel owner name happened to matched to the system's name. To eliminate these errors, we calculated the distance between the matched parcel and the mean center of that system's SDWIS locations. If a parcel with more than 5 miles away from the SDWIS locations, the service area was deleted. Approximately 1,500 service areas were deleted using this method.

The remaining parcels were manually checked for errors. Any PWSs with location errors were manually fixed if possible or deleted. In total, 77,470 PWS were matched out of a universe of 93,558 (83%).